

The Complete One-Gap Spectrum of the Degree-One Lamé Hamiltonian

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Abstract

We prove the complete real-line spectrum of the degree-one Lamé operator in a fixed Jacobi convention. Three explicit band-edge modes, a skew third-order commuting operator, and its cubic algebraic relation yield two bands, one finite open gap, and pure absolute continuity. The free-modulus and soliton limits, threshold resonance, translations, and all real-convention boundaries are closed explicitly.

1 Operator convention and band-edge modes

Fix $m = k^2 \in (0, 1)$ and write $K = K(m)$. On $L^2(\mathbb{R})$ let

$$H_m = -D^2 + u(x), \quad D = \frac{d}{dx}, \quad u(x) = 2m \operatorname{sn}^2(x \mid m), \quad (1)$$

with domain $H^2(\mathbb{R})$. The real potential is smooth, bounded, and has period $2K$. Hence the stated realization is self-adjoint and bounded below. All Floquet multipliers below are measured over $2K$.

The Jacobi differential identities are

$$\operatorname{sn}'' = -(1+m)\operatorname{sn} + 2m\operatorname{sn}^3, \quad (2)$$

$$\operatorname{cn}'' = (2m\operatorname{sn}^2 - 1)\operatorname{cn}, \quad \operatorname{dn}'' = (2m\operatorname{sn}^2 - m)\operatorname{dn}. \quad (3)$$

Consequently

$$H_m \operatorname{dn} = m \operatorname{dn}, \quad H_m \operatorname{cn} = \operatorname{cn}, \quad H_m \operatorname{sn} = (1+m)\operatorname{sn}. \quad (4)$$

Over one potential period, dn is periodic while cn and sn are antiperiodic. These formulas identify three edges, but by themselves do not prove that all higher gaps close.

2 Commuting operator and spectral curve

Set

$$A = -4D^3 + \{6u - 4(1+m)\}D + 3u'. \quad (5)$$

For $\theta \in [-\pi, \pi)$ define

$$H_\theta^j = \{f \in H^j([0, 2K]) : f^{(\ell)}(2K) = e^{i\theta} f^{(\ell)}(0), \ 0 \leq \ell < j\}.$$

The closed fiber realization of (5) has

$$D(A_\theta) = H_\theta^3([0, 2K]). \quad (6)$$

The coefficient of D has derivative $6u'$. Since all coefficients are $2K$ -periodic, every endpoint product in the Green boundary form cancels: the common phase $e^{i\theta}$ cancels its conjugate. Three integrations by parts therefore show $A_\theta^* = -A_\theta$; equivalently, the adjoint boundary form imposes the same three quasi-periodic conditions. Direct use of (2) gives the two identities

$$u'' = 3u^2 - 4(1+m)u + 4m, \quad (7)$$

$$(u')^2 = 2u^3 - 4(1+m)u^2 + 8mu. \quad (8)$$

Differentiating (7) and expanding differential-operator products, with (8) for the zeroth-order remainder, proves

$$[A, H_m] = 0, \quad A^2 = -16(H_m - m)(H_m - 1)(H_m - 1 - m). \quad (9)$$

The first integral is essential here; the stationary second-order equation alone leaves an arbitrary integration constant and does not prove the second identity in (9).

Theorem 1 (complete Lamé spectrum). *For every $0 < m < 1$,*

$$\sigma(H_m) = [m, 1] \cup [1 + m, \infty), \quad (10)$$

and the spectrum is purely absolutely continuous. Thus $(1, 1 + m)$ is the only finite open gap and every higher folded-zone gap is closed.

Proof. Let

$$R(E) = (E - m)(E - 1)(E - 1 - m).$$

Floquet decomposition realizes H_m as the direct integral of self-adjoint quasi-periodic operators with domain H_θ^2 on $[0, 2K]$. The operator A_θ has domain (6) and is skew-adjoint there. A fiber eigenfunction is smooth by its second-order ODE, lies in H_θ^3 , and is therefore in $D(A_\theta)$. If $H_m\psi = E\psi$ in a fiber, then (9) gives

$$-\|A\psi\|^2 = -16R(E)\|\psi\|^2.$$

Therefore a spectral energy must satisfy $R(E) \geq 0$.

Conversely, suppose $R(E) > 0$. On the two-dimensional solution space of $H_m\psi = E\psi$, equation (9) gives distinct eigenvalues $\pm 4i\sqrt{R(E)}$ for A . Because A commutes with period monodromy, its one-dimensional eigenspaces are Bloch lines. Real coefficients make the two Bloch solutions complex conjugates. If one has multiplier λ , the other has multiplier $\bar{\lambda}$. Their Wronskian is nonzero and constant; translating it by $2K$ multiplies it by $\lambda\bar{\lambda}$. Hence $|\lambda| = 1$, placing E in the spectrum. The three roots of R follow by closure and agree with (4). The condition $R(E) \geq 0$ is exactly (10).

For a real one-dimensional periodic Schrödinger operator, Floquet fibers have analytic nonconstant band functions; the direct integral therefore has neither eigenvalues nor singular-continuous spectrum. Its spectrum is purely absolutely continuous [4]. Since the second allowed interval extends from $1 + m$ to infinity without another excluded chamber, all higher folded-zone contacts are closed. This is an analytic completeness proof, not a finite coefficient inference. $\square$

The argument through (9) is the *commuting spectral-curve owner* of revision round zero.

3 Endpoint and equivalence atlas

At $m = 0$, equation (1) becomes the free operator $-D^2$ with spectrum $[0, \infty)$; the finite gap has closed. This face also explains why the infinitely many folded free contacts are not open gaps.

As $m \uparrow 1$, one has $K(m) \rightarrow \infty$ and, locally on the line,

$$u(x) \longrightarrow 2 \tanh^2 x = 2 - 2 \operatorname{sech}^2 x.$$

The lower band $[m, 1]$ collapses to the L^2 level 1, and the upper band tends to $[2, \infty)$. For the unshifted potential $-2 \operatorname{sech}^2 x$, the zero solution is a threshold resonance, not an additional L^2 eigenstate; after adding 2 it sits at threshold 2.

Changing k to $-k$ leaves $m = k^2$ and the operator unchanged. A spatial translation is unitary conjugacy and changes no band. Complex m , or real $m \notin [0, 1]$, lies outside the frozen real self-adjoint Jacobi convention. These statements are the *modulus boundary owner* added in revision round one.

Source boundary

This paper is a source-local reconstruction, not a priority claim. Ince's papers own the periodic Lamé analysis; NIST DLMF supplies an authoritative identity index; Kuchment supplies the general Floquet direct-integral and periodic absolutely-continuous spectral theorem. The checked JSON grid and release gates are new implementation receipts and are not attributed to those sources.

References

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